

Data Flow Diagram

Date	01 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI Based Crop Recommendation System
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data flows through the Human Development Index Prediction System. It shows the interaction between the user, the Flask web application, the machine learning model, and the prediction result. The DFD represents how user inputs are processed to generate the predicted Human Development Index (HDI) category.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Represents the Farmer/User who enters soil and environmental data and receives the recommended crop.
Rectangle with numbered header	Process	The Crop Recommendation Process that accepts user inputs, processes them using the trained machine learning model, and predicts the most suitable crop.
Rectangle (solid fill, no number)	Data Store	Stores the Crop Dataset, trained ML model (.pkl) , and other agricultural data used for prediction.
Labeled arrow	Data Flow	Shows the movement of data such as soil parameters (N, P, K, temperature, humidity, pH, rainfall) from the user to the system and the predicted crop back to the user.

Data Flow Diagram:

Data Flow Diagram: OptiCrop – AI Based Crop Recommendation System

